WEEK 8 REPORT

Interactive Road Accident Dashboard

Streamlit Development, Filtering and Visual Storytelling

Project	Road Accident Analysis
Team	DSV Team 14
Project stage	Week 8 – Interactive dashboard development
Dashboard framework	Python, Streamlit, Pandas and Plotly

Executive Summary

Week 8 focuses on turning the project's cleaned accident records and earlier exploratory analyses into an interactive Streamlit dashboard. The dashboard design brings together date and category filters, summary metrics, temporal charts, severity and condition comparisons, a location view, a record table, and a data-quality summary. The interface is intended to help users explore the records and trace patterns across the dimensions studied in earlier weeks.

Implementation status: The dashboard code and run instructions were prepared during the Week 8 work. The app still needs to be launched and its filters, charts, and file loading checked on the project computer. This report therefore documents the dashboard design and planned analysis, not verified visual findings or conclusions.

1. Introduction

The earlier project stages established the road-safety question, dataset representation, cleaning approach, exploratory analysis, feature engineering, and dimensionality-reduction work. Week 8 adds an interactive presentation layer so a viewer can adjust selections and inspect the corresponding records and summaries. The dashboard supports data storytelling by linking a question, a visual comparison, and a cautious interpretation.

2. Objectives

• Create a browser-based dashboard using Streamlit.

• Allow users to filter records by date, severity, weekday, road and condition codes, district, speed limit, vehicle count, casualty count, and hour.

• Present more than five visualization types, including a line chart, bar charts, histogram, scatter plot, heatmap, pie chart, and map.

• Show summary metrics and allow the filtered records to be inspected and downloaded.

• Display basic data-quality information and clearly communicate limits in category-code interpretation.

3. Dataset and Data Preparation

The supplied Week 3 cleaned file contains 59,999 records and 30 columns. It includes accident identifiers, date and time, severity, number of vehicles, number of casualties, road and condition fields, local-authority fields, and latitude/longitude. The file is gzip-compressed even though its filename ends in .csv; the dashboard loader therefore needs to read it with gzip compression enabled.

The app parses the date and time fields to derive month and hour for the temporal charts. Records that cannot be parsed as dates are excluded from the date-filtered view. The app computes summaries and charts from the filtered records while keeping the source data intact.

Data consistency note: The project framing refers to India, while some dataset field names and coding conventions require provenance confirmation. The supplied table has no State or Vehicle_Type field. Coordinate coverage and field meanings should be verified before presenting geographic or category labels as confirmed locations or descriptive names.

4. Dashboard Structure and Features

Section	User question / feature	Elements
Overview	What is the filtered record set like?	Record and casualty totals; average vehicles and casualties; severity-code share; vehicle and casualty histograms; vehicles-versus-casualties scatter plot.
Time patterns	When do the records occur?	Monthly trend line; weekday and hour bars; hourly record counts by severity code.
Road and conditions	How do records vary across recorded road and environment categories?	Bars for weather, light, road surface, road type, and urban/rural codes; grouped severity comparison; numerical correlation heatmap.
Locations	Where are coordinates available?	Map of records with latitude and longitude; top district-code counts.
Records and quality	What records and data issues underlie the charts?	Filtered record table; CSV download; missing-value summary; exact duplicate-row and duplicate-ID counts.

5. Filters and Interactivity

The sidebar is designed to let users select a date range and one or more available values for severity, weekday, road type, weather, light, road surface, urban/rural area, and district. Range sliders narrow records by speed limit, number of vehicles, and number of casualties; an hour slider filters by time of day. All dashboard sections use the resulting filtered dataset. Users can also download that filtered subset as a CSV file.

6. Interpretation and Data Safeguards

Several fields are stored as numeric category codes. Until a verified codebook is available, the dashboard labels these values as codes instead of assigning meanings such as “fatal”, “rain”, or a road-type name. Count differences and correlations are descriptive associations; the dashboard does not establish that a condition caused an accident. The map displays recorded coordinates and should be interpreted only after the dataset’s geographic provenance has been confirmed.

The Week 7 report identified blank rows that distorted PCA in its raw-data workflow. The supplied cleaned Week 3 file is a separate input and was observed to contain 59,999 populated rows. Any future PCA display should be rerun on the selected cleaned data and validated before reporting component percentages or conclusions.

7. Validation Plan and Limitations

Before demonstration, the team should confirm that the app launches from its project folder, reads the compressed CSV, displays the expected record count, and updates every chart and metric when filters change. The team should also check date parsing, missing coordinates, empty filter selections, CSV download behavior, and chart labels. Dashboard execution has not yet been confirmed in this report, so no chart-specific findings are stated.

Current limitations include dependence on a local CSV file, category codes without a confirmed lookup table, uncertain alignment between project framing and some dataset fields, and no validated PCA panel in the dashboard code.

8. Conclusion and Next Steps

Week 8 defines an interactive dashboard that brings the project's temporal, severity, road-condition, location, and data-quality views into one interface. The next steps are to run the app, resolve any setup or data-loading errors, verify the filters and visual outputs, confirm the dataset's codebook and geographic interpretation, and add short evidence-based takeaways after inspecting the charts. Once those checks are complete, the dashboard can support the project's final visual storytelling and demonstration.

References to Earlier Project Work

• Week 1 – India Road Accident Analysis: Raw Dataset, Abstract and Project Presentation.

• Week 2 – Dataset Representation and Initial Exploration.

• Week 3 – The Cleaning Sprint: Missing Values and Outlier Detection; supplied cleaned dataset.

• Week 4 – Exploratory Data Analysis of Road Accidents.

• Week 6 – Feature Engineering on a Road Accident Dataset.

• Week 7 – Dimensionality Reduction: Feature Selection, PCA and Variance Visualisation.

• Data-Viz Hack-a-Bit (2026–2027) Design Competition Rule Book.